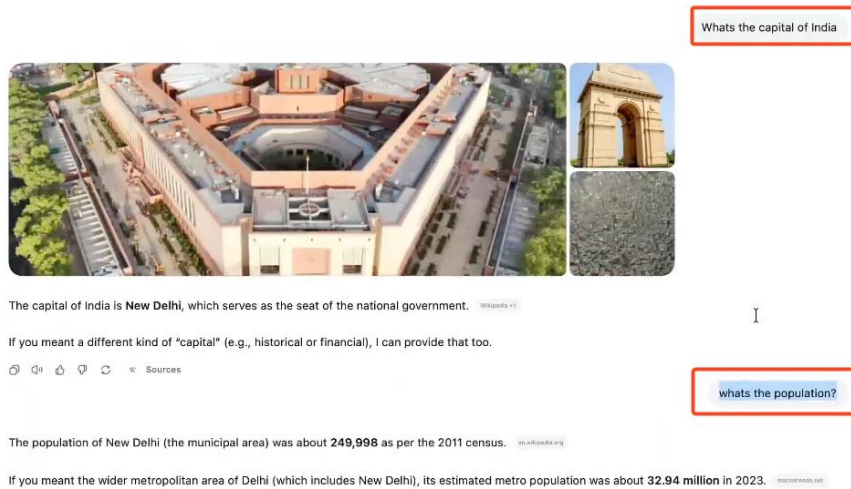


Understanding LLM Conversation Memory in Simple Terms



The above diagram explains how conversation history works when interacting with an LLM (Large Language Model).

Let us understand this step by step in simple language.

1. Example Conversation

Imagine you ask the following questions:

Message 1:

"What is the capital of India?"

Response:

"Delhi"

Immediately after that, you ask:

Message 2:

"What is the population?"

The system correctly understands that you are still talking about India and returns the population of India.

Then you ask:

Message 3:

"Who is the CM?"

Again, it understands the context and gives the correct answer.

2. What Is Actually Happening?

The main question is:

How does the system remember the previous question?

Is the LLM storing this information permanently?

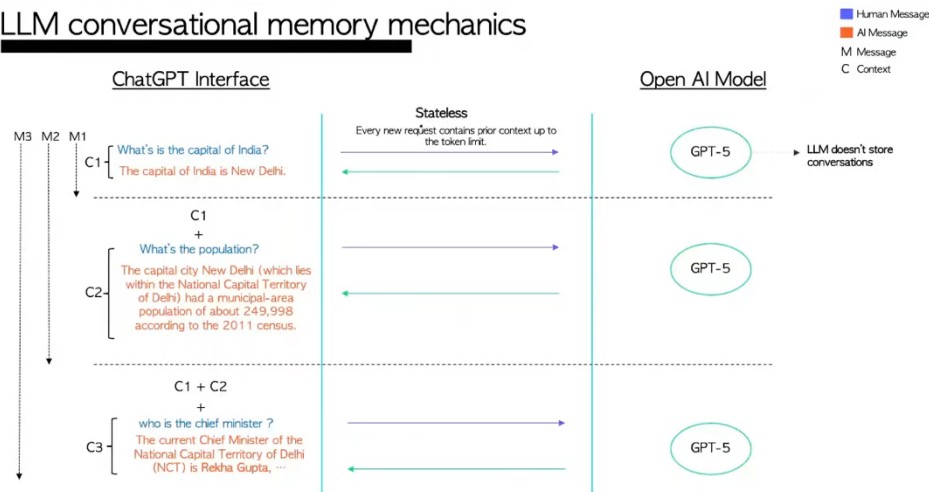
Is it being injected with knowledge?

The answer is:

LLMs do not permanently store your conversation data.

They do not remember users in the way humans do.

LLM conversational memory mechanics



3. How Conversation Memory Works

The memory does not live inside the LLM itself.

Instead, the application (such as ChatGPT, Gemini, or other AI platforms) manages the conversation history.

Here is what actually happens:

1. Every time you send a message, the application collects:
 - Your previous messages
 - The assistant's previous responses
2. It sends the entire conversation history again to the LLM along with your new question.
3. The LLM reads the full context and generates a response.

So technically:

The LLM is not remembering.

It is being given the previous conversation again every time.

4. Who Stores the Conversation?

The conversation history is stored by:

- The application (for example, ChatGPT platform)
- In temporary storage or cache
- For a limited time (may vary by company)

This is called conversation memory.

The storage duration may depend on:

- Company policies
- User settings
- System configuration

5. How the Second Question Works

Let us break down the second question:

Step 1:

User: "What is the capital of India?"

Assistant: "Delhi"

Step 2:

User: "What is the population?"

What the system actually sends to the LLM:

Full Context:

- User: What is the capital of India?
- Assistant: Delhi
- User: What is the population?

Because the word "India" appears in earlier context, the LLM understands that "population" refers to India.

It is not memory in the human sense.

It is context being re-sent.

6. Is the LLM Being Injected with Information?

Not exactly.

The LLM is not updated or trained in real time with your question.

Instead:

- It already has general knowledge from training.
- The conversation history is passed as input.
- It generates a response based on the input context.

No permanent storage happens inside the model for your session.

7. Important Clarification

LLMs:

- Do not permanently store personal conversation data.
- Do not learn from your session automatically in real time.
- Only process what is provided in the current request.

Applications:

- Manage session history.
- Cache conversation temporarily.
- Send context back to the model.

8. Simple Summary

When you ask multiple related questions:

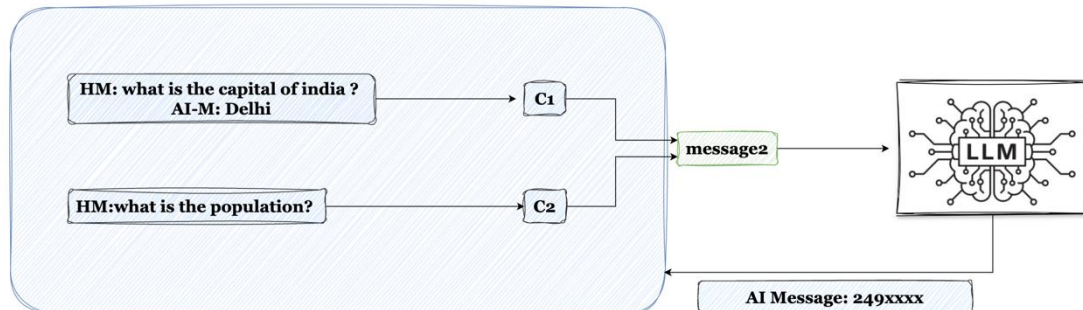
It looks like the model remembers.

But in reality:

- The application stores the conversation.
- The full conversation is sent again with each new message.
- The LLM reads the entire context and responds accordingly.

So the memory is at the application level, not inside the LLM.

how it remembers delhi ?



Final Understanding

LLM = Intelligent processing engine

Application = Manages conversation history

Context = Entire previous conversation sent again each time

This is how conversational AI systems maintain continuity in discussions.